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Multifaceted Syndromic Surveillance in a Public Health Department Using the Early Aberration Reporting System

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Local health departments concerned with early detection of potential terrorist threats are beginning to explore novel approaches to syndromic surveillance. Using the Early Aberration Reporting System (EARS) developed by the Centers for Disease Control and Prevention, a metropolitan health department in Tennessee and five community partners have agreed to exchange data in order to implement a multifaceted syndromic surveillance system for early detection of a biological attack. This article describes how we used EARS as the foundation for implementing a surveillance system that encompasses a rich variety of data sources. We address technical requirements for operating EARS, recommend staffing and training prerequisites, describe the involvement of our data partners, and provide details related to data transfer and analysis, review, and response protocol. Other health departments may find this information useful as a general model for implementing EARS-based syndromic surveillance systems in their own jurisdictions.

KEY WORDS: bioterrorism, public health surveillance, syndrome

Early detection of unusual patterns of health events as a component of bioterrorism surveillance is quickly becoming an essential function of local public health departments. According to the Centers for Disease Control and Prevention (CDC), initial detection of a covert biological attack will most likely occur at a local level.¹ Furthermore, outbreaks of infectious diseases, whether naturally occurring or intentional, often initially present as nonspecific syndromes in a va-

riety of health care settings.² Local public health departments can partner with other community health care agencies to establish novel surveillance systems for the early recognition of unusual patterns of nonspecific syndromes that may signal the onset of a biological attack or an emerging infectious disease.

The CDC has developed an analysis system to rapidly identify unusual patterns of health-related events. The Early Aberration Reporting System (EARS),³ which operates on a SAS⁴ platform, uses aberration detection models to identify changes in the frequency or distribution of health events when compared to historical data.⁵ A statistical quality control method called Cumulative Sums (CUSUM) detects shifts in the current daily count of a particular syndrome away from a historical mean.^{6,7} Variations of this system have been used for special events such as the Super Bowl in Tampa, Florida, and in the aftermath of the World Trade Center disaster.⁸

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The health department in Knox County, Tennessee, has implemented a multifaceted syndromic surveillance system that monitors an array of syndromes from six different community partners, including a 911 emergency call center, three hospital emergency departments (EDs), a large primary care medical group with more than 150 physicians, and the local public school system consisting of 86 schools throughout the county. Table 1 summarizes the syndromes monitored from each data partner.

Each agency electronically transmits (via e-mail or file transfer protocol [FTP]) data files that pertain to health events that occurred the previous day to the health department. Health department epidemiologists analyze the data with EARS and determine the need for follow-up on identified aberrations. Public health departments and community health agencies in other areas can begin to establish similar surveillance systems in their own geographic regions. In fact, a number of areas have already established surveillance systems for bioterrorism that partner with EDs, 911 emergency call centers, or other data partners.⁹ Many other systems focus on a specific data source for bioterrorism surveillance (eg, ED encounter data). However, we feel that the Knox County system is relatively unique in its integration of a rich variety of data sources into a multifaceted bioterrorism surveillance system.

This article describes how we implemented EARS with several different data sources in Knox County.

We describe the technical requirements necessary for a health department to implement EARS, recommend personnel and training prerequisites, and describe some specifics related to each collaborator’s involvement in developing our system. Finally, we describe our approach toward data transfer, analysis, review, and response. Other health departments may choose to use this as a general model for implementing an EARS-based surveillance system in their own geographic regions.

● Technical Requirements

Specific technical requirements for operating EARS can be found at the CDC-EARS Web site.¹⁰ Technical components of the system in Knox County are summarized in Table 2. Requirements include SAS 8.2 or higher running on a Windows 98 or higher operating system. SAS software may be available to local public health departments who qualify through a CDC grant.¹¹ The SAS components required to run EARS include BASE SAS, SAS/GRAPH, SAS STAT, and ACCESS to PC File Formats.

Other technical requirements will depend upon the method(s) used to transfer data from a reporting source to the local health department (eg, electronic transmission via e-mail or FTP, data entry using Web-based technologies, or manual input from reports received

TABLE 1 ● Data sources for the Early Aberration Reporting System (EARS) in Knox County, Tennessee

	911 Emergency call center	Outpatient physicians practice (150 physicians)	Metropolitan emergency departments	Public school system (86 buildings)
Active surveillance established	August 2001	May 2002	August 2002	August 2002
EARS Version*	Historical and nonhistorical	Nonhistorical	Nonhistorical	Nonhistorical
Data structure	Call category	ICD-9 code	Electronic Tracking Board disposition diagnosis	Absenteeism count for each school building
Monitored syndromes†	Abdominal/back pain	Pneumonia (486)	Abdominal pain	Sick/unknown
	Breathing difficulty	Influenza (780.6)	Diarrhea	
	Chest pain/heart problems	Bronchitis (490)	Chest pain (chest pain EUD and chest pain noncardiac)‡	
		Gastroenteritis (558.9)	Flu/flu symptoms	
	Sick/unknown	Diarrhea (787.91)	Pneumonia	
		Hemoptysis (786.3)	Gastroenteritis	
		Dysphagia (787.2)	Respiratory problems‡ (respiratory abnormality, respiratory difficulty/pain, and shortness of breath)	
Stratification‡	None	County of residence	County of residence	Geographic quadrant

*Two versions of EARS are available: a historical version calculates the mean from 3 to 5 years’ worth of baseline data (not including the current year), and a nonhistorical version calculates the mean from 7 days of baseline data.
†Similar syndromes were combined into more general categories. The original nursing disposition diagnoses appear in parentheses after the general category. EUD = etiology undetermined.
‡Stratification in the EARS programs is optional. Stratification is useful for determining whether clusters of syndromes exist within specific geographic regions.

TABLE 2 • Recommended information technology and staff training infrastructure for Early Aberration Reporting System—based syndromic surveillance

	Description
Information technology	High-end Computer SAS v.8.2 or higher: BASE SAS, SAS/GRAPH, SAS/STAT, ACCESS to PC file formats Internet connectivity (secure network connection) FTP server and software, or other means of automating data transfer (optional) Geographic information system such as ArcView 9.0 (optional) Internet browser (Internet Explorer 5.0 or Netscape 4.7)
Personnel/training	Epidemiologist or other staff member with intermediate programming skills Information technology staff for consultation regarding file type, transfer, and data automation issues

by fax or telephone). Our system receives data via e-mail and FTP. Since EARS produces tabular and graphical output in HTML, a recent version of an Internet browser (Internet Explorer 5.0 or Netscape 4.7) is required to view output.

In addition to information technology concerns, specific data elements are required to successfully operate EARS.¹² Data sent for analysis should include at a minimum the information related to (1) an event of interest such as an ED chief complaint, diagnosis code, or number of absentees in a school; (2) a date or other time period variable such as a specific day, week, or month of the year; and (3) the count of the event. Other information, such as geographic or demographic data, included in the data extract can facilitate stratified analysis in EARS.

● **Staffing Requirements**

In Knox County, a health department epidemiologist with intermediate SAS programming skills has primary responsibility for operating EARS. These skills include being able to import a variety of file formats into SAS, deal with missing data, recode variables, and using character and date functions. In addition to the epidemiologist, the Knox County public health officer (PHO) and communicable disease control nurses are consulted regularly for assistance with interpretation and follow-up recommendations. Two other staff members with SAS and EARS training are available as backups. Health department staff with expertise in computer networking and data management assist with file transfer issues.

● **Data Partner Descriptions**

911 Call center

In 2000, we made the decision to partner with the 911 emergency call center as a possible data source for bioterrorism detection based on the descriptions of a system in New York City (NYC). The NYC health department had developed a system to detect surges in influenza-like illness, using ambulance dispatch data.¹³ A Knox County Health Department (KCHD) epidemiologist contacted personnel at the CDC who were developing programs and aberration detection methods that would eventually become EARS. Working with the CDC, the KCHD epidemiologist and PHO approached the 911 emergency call center to discuss the surveillance proposal and obtain their participation in the project. Our 911 call center first released 5 years worth of historical data for use as baseline data with EARS. Arrangements were also made to obtain current data that corresponded to calls that occurred during the previous 24-hour period, as described below.

As calls are made to the 911 call center, they are triaged by symptom type. Although the center assigns calls into about 16 different categories, our PHO and the State Epidemiologist determined that four of these call types would be worthwhile to monitor for possible bioterrorism or other outbreaks. These call categories include (1) “breathing difficulty,” (2) “abdominal pain,” (3) “chest pain,” and (4) “sick/other” (see Table 1). The 911 call center agreed to export data on the number of calls received in each of the four categories over the preceding 24 hours and now does this on a daily basis. These data files are manually e-mailed to the health department each morning.

Emergency departments

The KCHD currently collects data from three metropolitan EDs, two of which are part of the same health system. We began this component of the surveillance project in 2002, when an infection control professional working on a hospital bioterrorism task force at one of the EDs contacted the KCHD to discuss our surveillance activities related to bioterrorism preparedness. At the time, we had been using EARS for 911 emergency call surveillance for approximately 1 year; the infection control professional suggested that we pilot a similar project at in her hospital’s ED. Although all three EDs began participating in surveillance at different times, the discussion below applies to all participating EDs since each utilizes a similar system for collection of patient-related information.

We held a series of meetings to determine how the EDs captured patient-related data and whether this

information could be extracted and forwarded to the KCHD. The EDs utilize a McKesson¹⁴ electronic tracking board to capture information about patients as they present for treatment. This system keeps an electronic record for each ED patient. After patients are initially screened by triage and patient data are entered into the tracking system, a physician evaluates the patient and any relevant laboratory work is completed. On the basis of the physician's assessment, a registered nurse in the ED assigns a disposition diagnosis to the patient when they are released from the ED and enters the diagnosis into a disposition field in the McKesson system. The disposition diagnosis is selected from a preidentified drop-down list developed by hospital staff in consultation with an ED educator, and it includes nearly 150 descriptions from which triage nurses can choose.

The EDs provided the health department PHO with a master list of these diagnosis descriptions. As with the 911 call data, senior health department staff conferred on which disposition diagnoses to monitor (see Table 1). Using such a drop-down diagnostic list lends some degree of standardization and quality control to assessing the reason for a patient's visit to the ED. Many other systems that monitor ED encounter data use patient chief complaint, ICD-9 codes, or other available data for surveillance purposes.⁹

The EDs developed a means of extracting data from their tracking system, and we began to collect baseline data for use with EARS. Data files contain information related to the date/time of the patient's visit, the disposition diagnosis, age, race, sex, city, and zip code. Files for two of the EDs are now automatically transferred via FTP, and discussions are in progress to transition the remaining ED to FTP. Initially, files contained information on all of the available disposition diagnoses. However, to comply with local hospital interpretations of Health Information Portability and Accountability Act (HIPAA) regulations,¹⁵ two of the hospitals developed limited data use agreements that restricted the file content to syndromes specifically monitored by the health department. Confidential patient information is not included in the raw data. Limited data use agreements of this sort are becoming more common as hospitals seek to clarify what data they are exchanging, with whom, in what format, and under what provision of HIPAA the disclosures are made. Fortunately, HIPAA regulations allow for the release of confidential patient care data, including identifiers for public health purposes as permitted by state and local disease reporting statutes.

Primary care medical group

Soon after the first ED began participating in the surveillance project, a large primary care medical

group serving Knox and surrounding counties in East Tennessee contacted the Health Department to inquire how they could assist in bioterrorism preparedness. This group employs more than 150 physicians and is composed of approximately 34 clinics in East Tennessee.

Typically, ICD-9 codes pertaining to a patient's diagnosis are input into the medical group's computerized billing management system the same day that patients are seen. Data from the system were written to a delimited text file based on the date that patients were charged for service. In addition to the charge date, raw data also included ICD-9 codes, age, race, sex, and zip code.

To select syndromes, public health providers and the medical director for the medical group conferred on which ICD-9 codes to monitor. These include seven clinical diagnoses such as pneumonia, influenza, gastroenteritis, bronchitis, hemoptysis, dysphagia, and diarrhea (see Table 1). Currently, data files are transferred via FTP to the health department and automatically downloaded each morning.

School absenteeism

Unusually high absenteeism rates (ie, the percentage of students absent) in specific geographic areas of the county may reflect either a bioterrorism event or an infectious disease outbreak. Our Health Department had a previously established working relationship with the county school system. The KCHD PHO serves as medical consultant to the school health nursing program, and the KCHD epidemiologist had worked with the school's computer software system to create a productivity reporting process for the school health nurses. We initially began manually monitoring school absenteeism as a percentage of all students absent from the entire county school system, primarily to monitor the burden of influenza and other winter season illnesses. As the school system developed their computer-based data-collection system, they began to electronically gather their data on school absenteeism. Building upon the success of using EARS with 911 call data, we made the decision in conjunction with school staff to collect and analyze school absenteeism data by individual schools throughout the county. Reporting absenteeism data by individual schools facilitates geographic information system analysis by school zone, as described below in the "Review and Response" section.

Eighty-six individual schools report the number of students absent each day to a central administrative office. This information is compiled into one data file, which contains the number of students absent each day, the date, a school building code, and the number of students enrolled for that day. Although we initially relied

upon e-mail to transfer data files, we have recently transitioned to an automated process, using FTP.

● Data Transfer and Analysis

EARS is a suite of SAS programs that includes a top-level program and several subprograms that calculate aberration detection measures and generate HTML output. Two versions of EARS utilize different baselines for the mean and standard deviation of an event of interest (ie, a syndrome). A historical version utilizes 3 to 5 years of baseline data. In Knox County, only the data from the 911 call center are analyzed with the historical version. A nonhistorical version of EARS calculates the mean and standard deviation based on 7 days of baseline data and is used to analyze all participating data providers in Knox County. Data from the 911 call center are also analyzed with the nonhistorical version for comparative purposes. For either version, when the current count of an event exceeds the mean by 3 standard deviations, the event is flagged as an aberration.

The EARS top-level program requires modification for each data provider to specify the appropriate computer directories and filenames, the event (syndrome) codes, and formats (eg, titles for graphs and tables) for HTML output. This needs to be done only once for each data provider, and documentation for modifying the top-level program is provided by the CDC. Stratification is an option in the top-level program if sufficient information is included in the raw data files. In Knox County, several raw data files contain zip code information, and data are stratified for analysis according to county of residence (see Table 1).

Each morning, data partners transfer raw data files containing information pertaining to the previous 24-hour period to the health department either via e-mail or automatically via FTP (see Table 1). Files that are uploaded via FTP are scheduled to be automatically downloaded at a specific time (usually just after they are posted to the server, which occurs very early in the morning). Files sent via e-mail must be manually retrieved. Typically, files from all data partners are available and analyzed by about 9:30 am each weekday.

Prior to running EARS, raw data from each provider must be appended into a SAS dataset that contains sufficient baseline data for the version of EARS that will be used for that data partner's analysis (ie, at least 3 years of baseline data for the historical version and at least 7 days of baseline data for the nonhistorical version).

We have developed MS-DOS batch files that reference EARS top-level SAS programs for each data partner to automate importing the data files and completing

the analysis. This is accomplished by setting the batch file as a scheduled task in Windows 2000 Task Scheduler. Once the analysis is complete, EARS generates both tabular and graphical HTML output (Figure 1). We typically review tabular output that provides daily counts for each event under surveillance for up to the previous 30 days. This table is also automatically e-mailed to infection control professionals and the administrative official of the outpatient medical group for their review.

If an event is flagged as an aberration, a notation (ie, C1, C2, C3) appears in the tables and graphs. These notations signify different levels of sensitivity related to the CUSUM statistic. In brief, the CDC describes C1 as having "mild sensitivity," C2 as having "moderate sensitivity," and C3 as having "ultra sensitivity." A thorough explanation of these aberration detection measures appears elsewhere.³ In our experience, we have come to rely solely on the C1 statistic in our daily review.

● Review and Response

It is important to note that aberrations do not necessarily signify outbreaks and may or may not be of public health interest.⁵ The decision to follow up on aberrations is ultimately made by the PHO. HTML output from EARS for each data source is examined daily (see Figure 1). When an aberration occurs, the epidemiologist examines its magnitude (ie, number of visits for a specific syndrome), temporal trends (ie, consecutive days with the same syndrome flagged), whether multiple syndromes were flagged on the same day, and output from other local data sources to look for corroborating trends.

Flagged events may also be examined spatially using Geographic Information System software (ArcView 9.0¹⁶) if geographic data are available. For example, if flags are raised for absenteeism rates in specific schools, the percentage of students absent can be projected onto a map to determine whether schools with the highest absenteeism rates fall within a specific geographic area of the county (Figure 2). Our other data partners incorporate geographic information such as latitude and longitude or patient zip codes in the data files that they send to the KCHD. Counts of flagged events meeting the criteria outlined above are projected onto a map to determine any evidence of spatial clustering, such as the majority of cases occurring in a specific zip code area. We are currently exploring the utility of more formal cluster detection algorithms such as those available with ArcView's Spatial Analyst extension.¹⁷

If the PHO determines that follow-up is indicated, we will contact appropriate personnel at the data

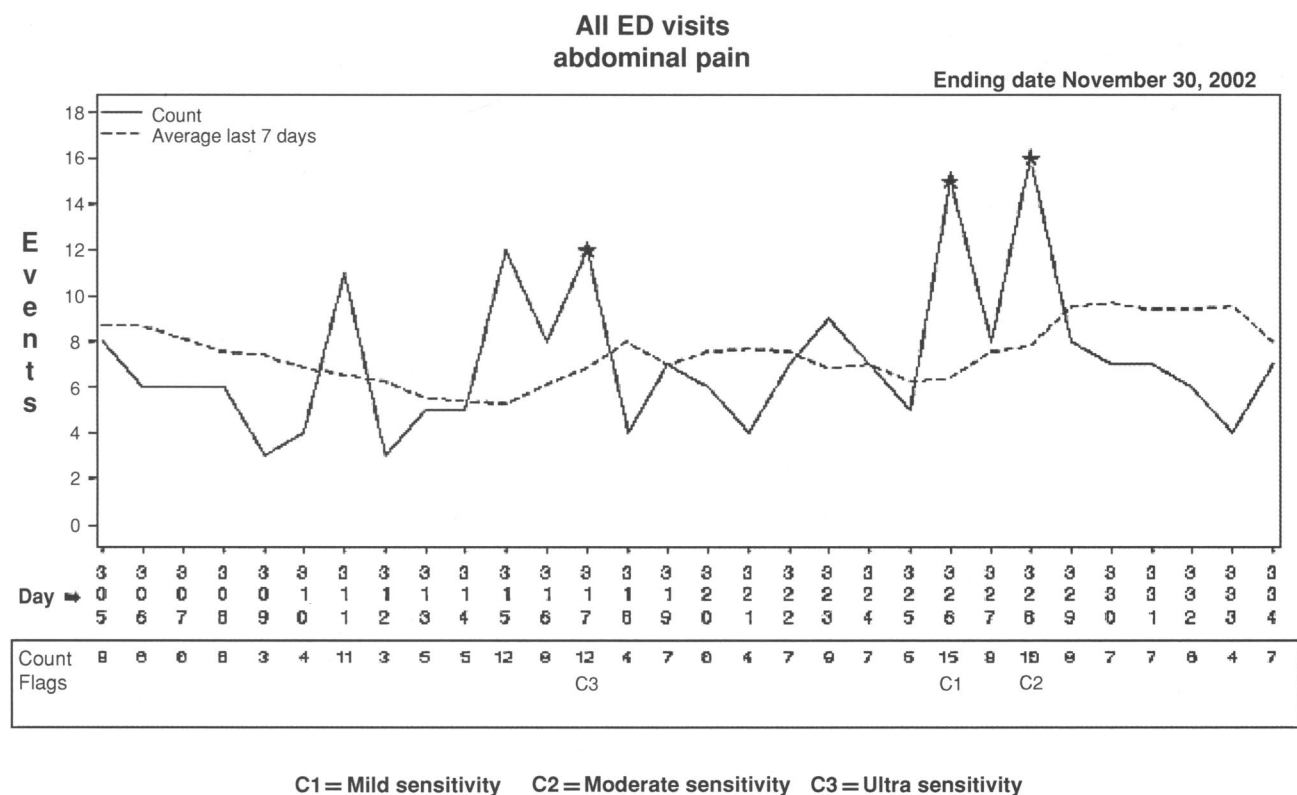


FIGURE 1. Graphical output from the EARS showing flagged aberrations in ED visits related to abdominal pain.

source to begin a joint preliminary epidemiological investigation that analyzes commonalities between patients with the same symptoms. Further action will be based on any findings from this preliminary investigation. Although we have conducted a number of secondary GIS and demographic analyses, the PHO has never had to require more rigorous investigative follow-up based on the outcome of these analyses.

When first implemented, review and analysis of data was limited to the work week. However, the KCHD has begun taking steps to facilitate weekend access to system output, data, and programs. For data partners who utilize FTP, weekend review of data is possible because file transfer and analysis occurs automatically via the scheduled tasks discussed in the previous section. HTML output is automatically e-mailed to relevant KCHD personnel for review over the weekend. However, access is currently limited to HTML output only; at this time we do not have access to data files and programs to conduct any follow-up analyses. To address the weekend access issue, the KCHD currently provides high-speed Internet service to an epidemiologist and backup personnel in their homes, and we are in the process of installing a virtual private network and software to enable secure remote access to the computer at the KCHD that houses all relevant data files and pro-

grams. Once in place, 7-day-per-week review and analysis of data will be possible.

● **Evaluation**

Syndromic surveillance systems for bioterrorism detection are relatively new and need to be evaluated. A draft framework for evaluating syndromic surveillance systems has been developed by the CDC.¹⁸ This framework will be used as a guide for the system in Knox County to examine both product and process from EARS. Output (product) will be examined in terms of specificity and sensitivity by focusing on comparisons between data sources, geographic location (urban vs rural), and EARS methodology (historical baselines vs nonhistorical baselines). Datasets for evaluating and testing EARS are available on the EARS Web site.¹⁰ Process evaluation will be directed toward EARS stakeholders, including state and local epidemiologists, PHOs, information technology specialists, and community collaborators that share data with the health departments.

This system has provided several benefits. Monitoring influenza-like illness from local EDs and the outpatient physician group has been helpful for predicting and modeling trends during influenza season. The physician group has also benefited from influenza

Percentage of elementary school students absent by school zone
Knox County, Tennessee

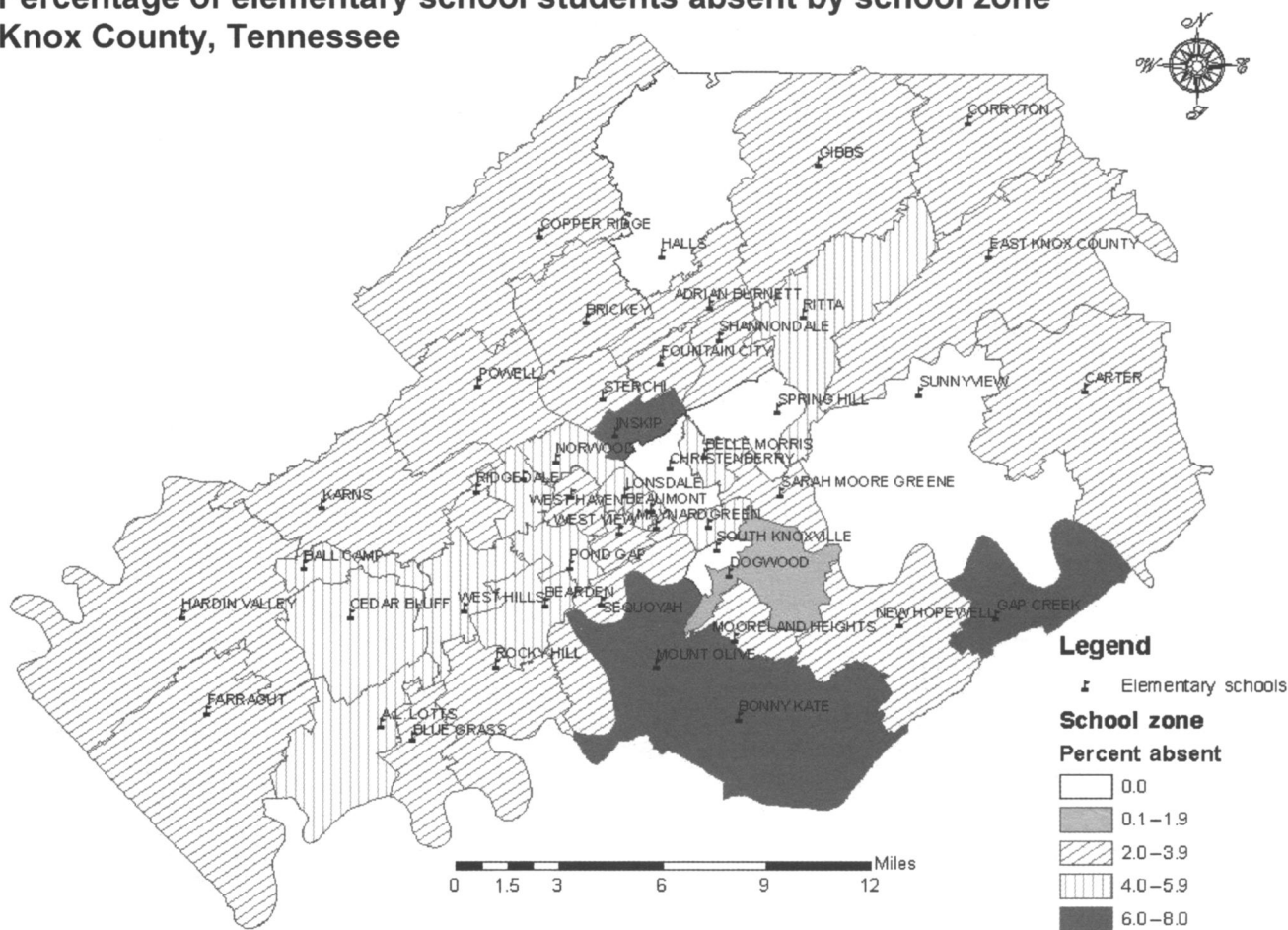


FIGURE 2. A sample map showing elementary school absenteeism rates.

monitoring since they now rely on electronic reporting of influenza cases as opposed to telephonic reporting. In addition, influenza counts coupled with information regarding school absenteeism rates have helped health department officials assist public schools in determining whether to close the school. Viral outbreaks have also been detected through the monitoring of individual school building absenteeism rates. Finally, developing this system has clearly helped forge new and strengthen existing relationships by engaging community partners in discussions related to bioterrorism preparedness and sharing relevant data.

● Limitations

This article presents an example of how a local public health department can develop a multifaceted EARS-based syndromic surveillance system with several community partners. Health departments and community agencies in other areas can use the information presented here as a general model. In fact, on the basis of

the experiences of the pilot project in Knox County, the Tennessee Department of Health has integrated syndromic surveillance systems as a component of normal surveillance activities to be conducted across the state.¹⁹ As these systems were developed in other parts of the state, some limitations became apparent as outlined below.

1. The Knox County system is based on community agencies that collect and track information electronically. Other health departments, particularly those in rural regions, may not have access to electronic data. However, to address this limitation, some rural health departments in Tennessee have developed systems that rely on facsimile transmissions of aggregate data.
2. Initial data transmission in Knox County utilized nonsecure e-mail to transfer data. Only two data partners still rely on e-mail to transfer files, but this is still a significant limitation since e-mail is less secure than other Web-based technologies for transferring information. In addition, the use of e-mail does not

permit automating data transfer and analysis. Other health departments may wish to explore a means of secure, automated data transmission prior to implementing ongoing surveillance. Utilizing a secure FTP or Web-based connection adequately protects data and facilitates automated data transfer; it is the current gold standard for secure data transmission.

3. Not all health departments will have the same resources to implement an EARS-based surveillance system as those identified in this article. The resources and personnel used to implement EARS in Knox County can be used as a general guide for other health departments that choose to conduct syndromic surveillance. In other areas, circumstances may warrant the use of alternative resources and personnel for developing these systems. For instance, smaller health departments without the infrastructure to implement FTP or secure e-mail may wish to consult with their local government offices regarding available information technology resources.

● Conclusion

In this article we describe how a local health department has used EARS as a foundation for developing a multifaceted syndromic surveillance system for bioterrorism and other disease outbreaks. We have found that with the appropriate technical and personnel infrastructure, a local public health department can readily engage a variety of community stakeholders in data-sharing relationships to implement an automated bioterrorism surveillance system. The EARS system is very flexible in that the top-level program is easily modified to adjust to a range of data scenarios (eg, chief complaints, ICD-9 codes, absenteeism data, etc), and HTML output is easily shared among health department staff and data partners. Intra-agency collaboration in this process may better prepare communities to detect and respond to possible bioterrorism events, enhance the surveillance and response capacities for other health-related events in the community, and strengthen existing collaborative relationships among community partners.

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